

XINGTAI, China

WMO: 537980

Lat: 37.1808N

Lon: 114.3589E

Elev: 604

StdP: 14.38

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	21.5	-11.1	3.1	32.6	-6.6	3.9	33.6	16.1	35.4	13.5	34.9	3.3	20	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	96.6	72.1	93.8	72.9	91.4	73.3	81.2	89.0	79.8	87.1	78.3	85.4	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	154.5	86.0	77.7	147.3	85.1	76.2	139.8	83.9	45.6	89.5	44.1	87.2	42.5	85.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
16.2	12.9	10.7	DB	14.4	101.7	2.8	2.6	12.4	103.6	10.8	105.2	9.2	106.7	7.1	108.6	86.4
			WB	12.0	83.4	2.7	1.2	10.0	84.3	8.4	84.9	6.9	85.6	4.9	86.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.9	31.2	37.7	49.7	61.4	71.8	79.9	82.0	79.4	71.6	60.7	45.6	34.6	
	DBStd	18.85	5.60	6.83	7.88	7.14	6.08	5.46	4.73	4.58	5.48	6.32	7.69	5.60	
	HDD50	1689	581	347	101	6	0	0	0	0	0	4	173	477	
	HDD65	4149	1046	764	479	150	15	0	0	0	11	160	582	942	
	CDD50	4945	0	3	91	347	677	896	993	912	649	336	41	0	
	CDD65	1932	0	0	4	41	227	446	528	448	211	27	0	0	
	CDH74	19661	0	1	38	348	2008	5040	6145	4438	1454	184	5	0	
	CDH80	8137	0	0	7	79	710	2401	2832	1688	392	28	0	0	
Wind	WSAvg	4.5	3.9	4.4	5.5	5.7	5.6	5.1	4.4	4.1	4.0	3.8	3.7	3.7	
Precipitation	PrecAvg	20.5	0.2	0.3	0.5	1.0	1.6	2.2	5.9	4.8	2.2	1.0	0.7	0.2	
	PrecMax	31.7	0.7	1.7	2.2	2.1	4.3	4.4	14.6	13.8	4.3	4.0	2.6	0.9	
	PrecMin	13.7	0.0	0.0	0.0	0.2	0.2	0.5	1.9	1.3	0.2	0.1	0.0	0.0	
	PrecStd	4.8	0.2	0.4	0.6	0.6	1.0	0.9	3.1	2.8	1.2	0.9	0.8	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.1	62.2	78.8	87.6	96.1	101.1	100.3	95.6	90.3	83.5	72.2	55.5	
		MCWB	39.0	45.3	54.5	61.7	64.9	69.8	74.4	78.5	72.0	61.5	52.1	40.1	
	2%	DB	47.4	56.9	72.3	81.8	90.0	97.0	96.5	92.3	87.1	78.8	65.9	50.6	
		MCWB	35.1	42.4	53.1	61.3	65.2	69.1	74.8	77.4	69.3	59.1	50.4	37.9	
	5%	DB	43.8	53.0	67.4	77.9	87.0	94.0	93.7	90.1	84.5	75.2	61.6	46.8	
		MCWB	33.2	40.0	50.4	59.3	64.4	68.9	76.0	76.4	68.1	58.2	48.5	36.1	
	10%	DB	40.4	49.1	62.7	74.2	83.8	91.3	91.3	88.0	81.8	71.8	57.4	43.4	
		MCWB	31.3	38.2	47.9	57.7	63.9	69.2	76.2	75.7	67.4	57.6	46.4	34.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	48.2	59.8	69.1	72.8	77.6	83.3	82.6	77.9	66.8	55.4	42.0
			MCDB	51.9	57.8	71.1	79.7	83.6	89.8	91.8	90.7	85.4	75.9	66.3	51.6
2%		WB	36.1	44.1	54.9	65.5	70.8	75.3	81.6	81.1	74.6	64.1	52.6	39.2	
		MCDB	45.5	53.3	68.7	76.0	82.3	87.5	89.6	88.6	81.9	71.7	61.5	46.9	
5%		WB	33.8	41.4	51.8	62.6	68.9	73.8	80.2	79.8	72.0	62.1	50.3	37.4	
		MCDB	42.3	50.4	64.4	73.4	81.0	85.7	87.9	86.7	79.7	69.9	59.1	44.9	
10%		WB	31.9	39.0	49.1	60.1	67.0	72.4	79.0	78.2	70.1	60.2	48.2	35.5	
		MCDB	39.3	48.3	60.8	70.4	79.0	84.3	86.3	84.8	78.0	67.7	56.0	42.1	
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.2	16.3	17.1	17.3	17.3	13.8	13.1	15.3	16.2	14.8	12.7	
		MCDBR	18.9	20.9	22.6	23.1	22.8	22.2	18.1	16.1	19.1	21.9	21.8	19.0	
		MCWBR	11.4	11.9	11.6	10.9	10.0	6.6	5.9	6.2	7.0	8.7	10.8	10.9	
	5% WB	MCDBR	17.4	18.6	20.8	19.9	18.5	17.2	13.9	13.6	14.7	15.9	19.0	16.5	
		MCWBR	10.9	11.4	11.6	11.3	9.8	6.4	6.2	6.5	7.0	8.0	10.3	10.1	
Clear-Sky Solar Irradiance	taub	0.498	0.577	0.579	0.619	0.651	0.796	0.782	0.759	0.701	0.647	0.529	0.480		
	taud	1.811	1.671	1.694	1.651	1.635	1.439	1.506	1.528	1.583	1.635	1.795	1.868		
	Ebn at Noon	200	200	218	220	216	186	187	187	187	179	188	195		
	Edh at Noon	51	67	71	78	81	98	91	88	79	68	52	45		
All-Sky Solar Radiation	RadAvg	768	944	1374	1643	1817	1665	1466	1421	1233	1013	805	715		
	RadStd	83	145	131	123	145	139	93	140	120	118	133	81		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page